

Devi Ahilya University, Indore
Institute of Engineering & Technology

Class Test: I IT-A Engineering

1MERC3 Elements of Mechanical Engineering

Duration: 70 Min.

Max. Marks: 20

Note: Attempt all Questions. Draw neat sketches wherever required.

Q.1 What is Pattern. List different types of Pattern and Explain any one.

(5)

Q.2 Explain First law of thermodynamics (5)

Q.3 Define is Casting. Explain different properties of molding sand in casting process. (5)

130C

Q.4 0.2m³ of air at 4 bar and ~~1500~~¹³⁰⁰C is contained in a gas. A reversible adiabatic expansion takes place till the pressure fall to 1.02 bar. The gas is heated constant pressure till enthalpy increses by 72.5kJ. Calculate the work done. (5)

(1) what is principle of welding & types of welding process

(3) Explain arc-welding process.